Figure 10.1 **(**a**)** A supercooled Higgs field is one whose value gets trapped on the energy bowl’s high-energy plateau, like the frog on a bump. **(**b**)** Typically, a supercooled Higgs field will quickly find its way off the plateau and drop to a value with lower energy, like the frog’s jumping off the bump.

The second difference is that whereas Einstein carefully and arbitrarily chose the value of the cosmological constant—the amount of energy and negative pressure it contributed to each volume of space—so that its outward repulsive force would precisely balance the inward attractive force arising from the ordinary matter and radiation in the cosmos, Guth was able to estimate the energy and negative pressure contributed by the Higgs fields he and Tye had been studying. And the answer he found was more than 10000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000 (10^{100}) times larger than the value Einstein had chosen. This number is huge, obviously, and so the outward push supplied by the Higgs field’s repulsive gravity is *monumental* compared with what Einstein envisioned originally with the cosmological constant.

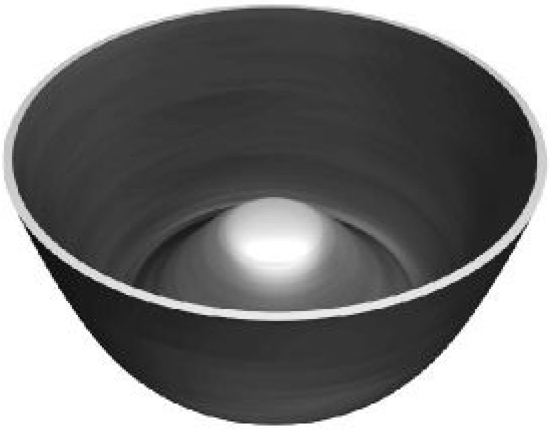

Figure 10.2 A smoother and more gradually sloping bump allows the Higgs field value to roll down to the zero-energy valley more easily and more uniformly throughout space.

Now, if we combine these two observations—that the Higgs field will stay on the plateau, in the high-energy, negative-pressure state, only for the briefest of instants, and that while it is on the plateau, the repulsive outward push it generates is enormous—what do we have? Well, as Guth